

# Math is Ingenuity

Lesson 10-4

**Name:**

**Date:**

**Class:**

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## TODAY'S OBJECTIVES

**Content Objective:** I can use ratios and models to explain how the Penny Farthing and gear-driven bicycles were ingenious solutions to a real problem.

**Language Objective:** I can explain a gear ratio using the words rotation, teeth, and 'for each \_\_\_\_, the wheel turns \_\_\_\_.'



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

ingenuity

rotation

gear

ratio

equivalent ratios



Tap any word to see its meaning and a picture.

1

Coming up with a different, sometimes unique solution is

2

One full turn of a wheel, gear, or pedal is a

3

A toothed wheel that passes motion through a chain is a

4

A comparison of two quantities, like pedal turns to wheel turns, is a

5

Ratios that make the same comparison with different numbers are

## Write About the Math

**How will your future vehicle be powered, and what math will be used in its design or use?**

*¿Con qué funcionará tu vehículo del futuro, y qué matemáticas se usarán en su diseño o su uso?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **ingenuity**  
*ingenio*

☐ **rotation**  
*rotación*

☐ **gear**  
*engranaje*

☐ **ratio**  
*razón*

☐ **equivalent ratios**  
*razones equivalentes*

**Model:** One pedal rotation makes one wheel rotation, so a bigger wheel's longer perimeter carries the bicycle farther with each pedal push. — A strong answer connects the 1-to-1 ratio to the wheel's perimeter, explaining a bigger wheel travels farther per pedal push. A weak answer just says 'bigger is faster' without naming the ratio or perimeter.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Describe a future mode of individual transportation. A complete answer says how it is \_\_\_\_, names a \_\_\_\_ (or rate) that shapes it, and explains where that math appears in the vehicle's \_\_\_\_.**
  - **My answer is \_\_\_\_ because \_\_\_\_.**  
*Mi respuesta es \_\_\_\_ porque \_\_\_\_.*
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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*

- **My evidence is** \_\_\_\_, **so** \_\_\_\_.

*Mi evidencia es* \_\_\_\_, *entonces* \_\_\_\_.

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## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.